

CLASSICAL VS DIRECTIONAL DO-NOTHING CONDITION

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1 Motivation

Let us consider an incompressible Navier-Stokes fluid flowing rightward in a channel. In order to run computational simulations on such a motion, the channel needs to be approximated with a finite-length channel. That means, taking a cutout geometry of the channel in question (Fig. 1). It is important to set the correct boundary conditions. In this project, we consider two boundary conditions: (a) the classical *do-nothing boundary condition* (CDN) and (b) the *directional do-nothing condition* (DDN) and compare. The DDN should show better numerical stability due to its "energy-stable" construction (see [1]).

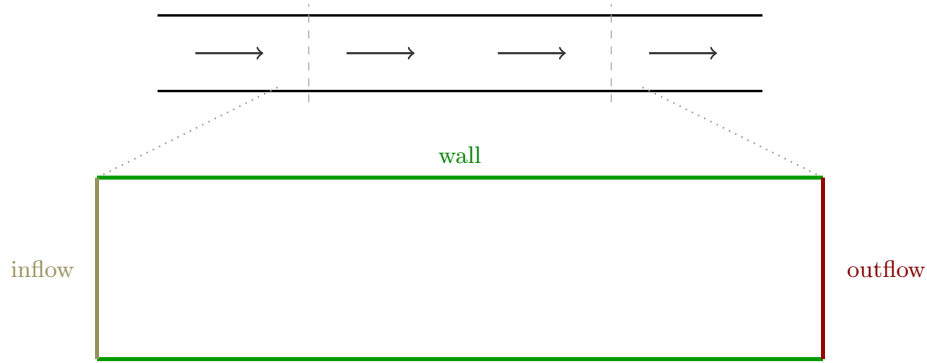


Figure 1: *Up:* Rightward flow in a channel. *Down:* Bounded approximation of a channel: wall (*green*), inflow (*yellow*), outflow (*red*).

2 Mathematical description of the flow

We consider a model described on domain Ω with a Lipschitz boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$ by the following. We solve for $(\mathbf{v}(t), p(t))$, where $\mathbf{v}$ is the velocity field, p is kinematic pressure and t is time, $t > 0$. We also use the notation $\tilde{\mathbb{T}} := -p\mathbb{I} + \nu\nabla\mathbf{v}$ for kinematic pseudo-stress tensor, $\mathbb{I}$ for identity matrix, ν for kinematic viscosity, $\mathbf{f}$ for a source term, $\mathbf{g}$ and $\mathbf{h}$ for boundary profiles, $\mathbf{v}_0$ for initial velocity, $\mathbf{n}$ for outer normal at boundary, $\mathbf{v}_r$ for a reference flow. We set the test space $V = W_{\Gamma_D, div}^{1,2}(\Omega) := \{\varphi \in [W^{1,2}(\Omega)]^d : \nabla \cdot \varphi = 0, \varphi = 0 \text{ on } \Gamma_D\}$.

2.1 CDN boundary condition

The governing equations with the CDN boundary condition read

$$\begin{aligned} \partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v} - \nabla \cdot \tilde{\mathbb{T}}(\mathbf{v}, p) &= \mathbf{f} && \text{in } \mathbb{R}^+ \times \Omega \\ \nabla \cdot \mathbf{v} &= 0 && \text{in } \mathbb{R}^+ \times \Omega \\ \mathbf{v} &= \mathbf{g} && \text{on } \mathbb{R}^+ \times \Gamma_D \\ \mathbf{n} \cdot \tilde{\mathbb{T}}(\mathbf{v}, p) &= \mathbf{0} && \text{on } \mathbb{R}^+ \times \Gamma_N \\ \mathbf{v}(0, \cdot) &= \mathbf{v}_0(\cdot) && \text{in } \Omega \end{aligned} \tag{1}$$

with the weak formulation being $\forall \varphi \in V$

$$\int_{\Omega} \partial_t \mathbf{v} \cdot \varphi \, dx + \int_{\Omega} (\mathbf{v} \cdot \nabla \mathbf{v}) \cdot \varphi \, dx + \nu \int_{\Omega} \nabla \mathbf{v} : \nabla \varphi \, dx = \int_{\Omega} \mathbf{f} \cdot \varphi \, dx \tag{2}$$

2.2 DDN condition

The governing equations with the DDN boundary condition read

$$\begin{aligned}
 \partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v} - \nabla \cdot \tilde{\mathbb{T}}(\mathbf{v}, p) &= \mathbf{f} & \text{in } \mathbb{R}^+ \times \Omega \\
 \nabla \cdot \mathbf{v} &= 0 & \text{in } \mathbb{R}^+ \times \Omega \\
 \mathbf{v} &= \mathbf{g} & \text{on } \mathbb{R}^+ \times \Gamma_D \\
 \mathbf{n} \cdot \tilde{\mathbb{T}}(\mathbf{v}, p) + \frac{1}{2}[\mathbf{v} \cdot \mathbf{n}]^-(\mathbf{v} - \mathbf{v}_r) &= \mathbf{h} & \text{on } \mathbb{R}^+ \times \Gamma_N \\
 \mathbf{v}(0, \cdot) &= \mathbf{v}_0(\cdot) & \text{in } \Omega
 \end{aligned} \tag{3}$$

with the weak formulation being $\forall \varphi \in V$

$$\int_{\Omega} \partial_t \mathbf{v} \cdot \varphi \, dx + \int_{\Omega} (\mathbf{v} \cdot \nabla \mathbf{v}) \cdot \varphi \, dx + \nu \int_{\Omega} \nabla \mathbf{v} : \nabla \varphi \, dx + \int_{\Gamma_N} \frac{1}{2}[\mathbf{v} \cdot \mathbf{n}]^-(\mathbf{v} - \mathbf{v}_r) \cdot \varphi \, dS = \int_{\Omega} \mathbf{f} \cdot \varphi \, dx + \int_{\Gamma_N} \mathbf{h} \cdot \varphi \, dS \tag{4}$$

where

$$[a]^- := \max(-a, 0) \quad \forall a.$$

2.3 Our problem

We consider a 2D square geometry (Fig. 2), where $(x, y) \in \Omega = (0, 1)^2$, with Dirichlet boundary condition on $\Gamma_D = \Gamma_{in} \cup \Gamma_{wall}$ read

$$\mathbf{g} = \begin{cases} \left(\frac{4V_{max}}{(0.4-0.1)^2}(y-0.1)(0.4-y), 0 \right) & \text{on } \Gamma_{in} = \{0\} \times (0.1, 0.4) \\ (0, 0) & \text{on } \Gamma_{wall} = \partial\Omega \setminus (\Gamma_{in} \cup \Gamma_{out}) \end{cases} \tag{5}$$

(which is a parabolic profile on inflow for maximal velocity magnitude V_{max} (we set $V_{max} = 1$) and no-slip on walls) and Neumann do-nothing condition on $\Gamma_N = \Gamma_{out} = 1 \times (0, 1)$. We also consider $\mathbf{f} = \mathbf{v}_r = \mathbf{h} = \mathbf{0}$.

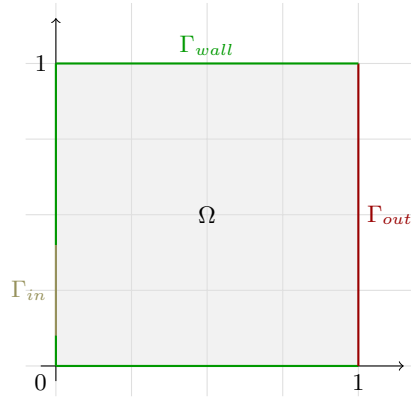


Figure 2: Square geometry.

3 Simulations

For a viscosity $\nu = 10^{-k}$, we compare the results using CDN and DNN. We set the following parameters:

$k = 3$	<i>viscosity</i>
$nx, ny = 20, 20$	<i>mesh resolution</i>
$dt = 0.01$	<i>time step</i>
$tmax = 20$	<i>number of time steps</i>
space discretization	Taylor-Hood P2/P1
time discretization	backward Euler
solver	Newton's line search

	convergence	time step	time
CDN	diverged	after 641	6.41s
DDN	converged	in 2000	20s

Table 1: Convergence of CDN and DDN simulations.

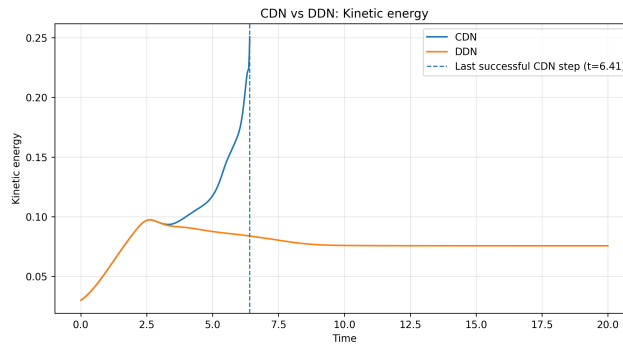
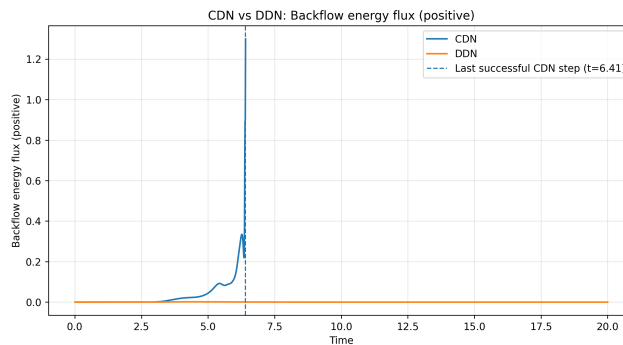
4 Results

Table 1 shows the differences between the CDN and DDN case regarding convergence.

Figure 3 shows that the energy in both simulations initially evolves in a similar manner. After approximately $t = 3$, the trajectories begin to diverge. In the case of CDN, the energy subsequently rises sharply, reaching a value of approximately 0.25 just before the solver fails. In contrast, for DDN, the energy gradually decreases after an initial rise, approaching a steady-state value of around 0.07–0.08. This observation aligns with the theoretical motivation behind DDN: limiting the energy contribution from backflow allows for better control over kinetic energy.

Figure 4 shows a significant increase in the energy contribution from backflow in the CDN simulation just before the solver fails. In the case of DDN, this contribution remains very small compared to CDN. This is interesting because DDN not only compensates for the boundary term in the energy balance but evidently also limits the energy flux associated with the backflow itself.

Lastly, Figure 5 helps explain the previous two. The return volume flux for the CDN begins to rise significantly, reaching approximately 0.3 around the time of failure. For the DDN, after an initial rise, it decreases and remains substantially lower. A connection is thus evident between the increasing return flux at the open boundary, the rise in kinetic energy, and the failure of the CDN solver.

**Figure 3:** Kinetic energy.**Figure 4:** Backflow energy flux.

5 Conclusion

In this project, we compared the classical do-nothing (CDN) and directional do-nothing (DDN) boundary conditions for unsteady incompressible Navier–Stokes flow in a two-dimensional square domain with a prescribed parabolic inlet profile.

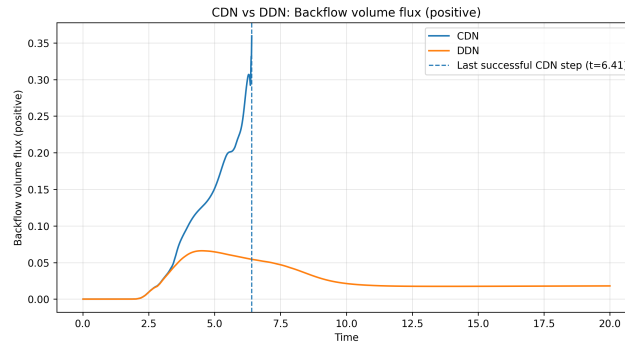


Figure 5: Backflow volume flux.

For the considered configuration with viscosity $\nu = 10^{-3}$, the CDN simulation failed to converge at approximately $t = 6.41$, whereas the DDN simulation successfully reached the final time $t = 20$.

The numerical results reveal substantial differences in the evolution of kinetic energy and backflow through the artificial outlet boundary. These observations are consistent with the theoretical motivation of the DDN condition, which removes the unfavorable contribution of backflow from the energy balance at the artificial outlet.

References

- [1] Malte Braack and Piotr Boguslaw Mucha. Directional do-nothing condition for the navier-stokes equations. *Journal of Computational Mathematics*, 32(5):507–521, June 2014.